

OCR Further Mathematics
Pure Core - Hyperbolic Functions
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a) You are given that $f(x) = 4 \sinh x + 3 \cosh x$.

Show that the curve $y = f(x)$ has no turning points.

[3]

(b) Determine the exact solution of the equation $f(x) = 5$.

[5]

2(a) In this question you must show detailed reasoning.

Find the values of A , B and C for which $\frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4} \equiv A + \frac{Bx + C}{x^3 + x^2 + 4x + 4}$.

[1]

(b)

Hence express $\frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4}$ using partial fractions.

[5]

(c)

Using your answer to part (b), determine $\int_0^2 \frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4} dx$ expressing your answer in the form $a + \ln b + c\pi$ where a is an integer, and b and c are both rational.

[4]

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3(a) In this question you must show detailed reasoning.

Using the definition of $\cosh x$ in terms of exponentials, show that $\cosh 2x \equiv 2\cosh^2 x - 1$.

[2]

(b) Solve the equation $\cosh 2x = 3\cosh x + 1$, giving all your answers in exact logarithmic form.

[6]



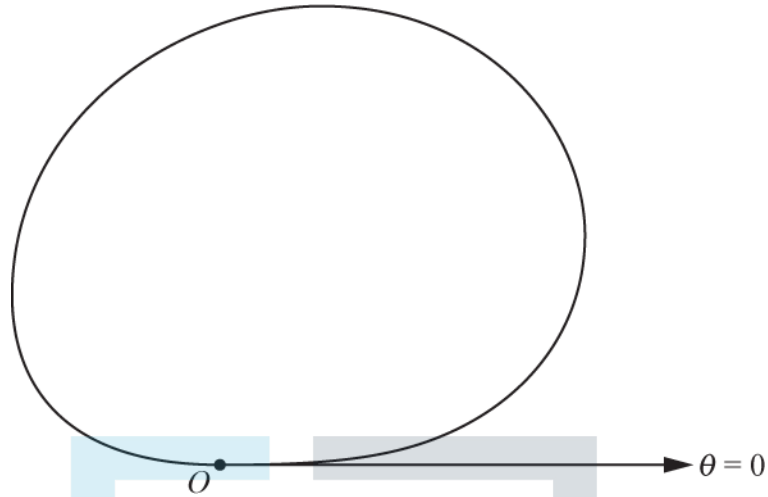
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4

The diagram below shows the curve $r = \sqrt{\sin\theta}e^{\frac{1}{3}\cos\theta}$ for $0 \leq \theta \leq \pi$.



Find the exact area enclosed by the curve.

[4]

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5 Let $f(x) = \sin^{-1}(x)$.

By writing $f(x)$ as $\sin^{-1}(x) \times 1$, determine the value of $\int_0^{\frac{1}{2}} f(x) dx$. Give your answer in exact form.

[3]



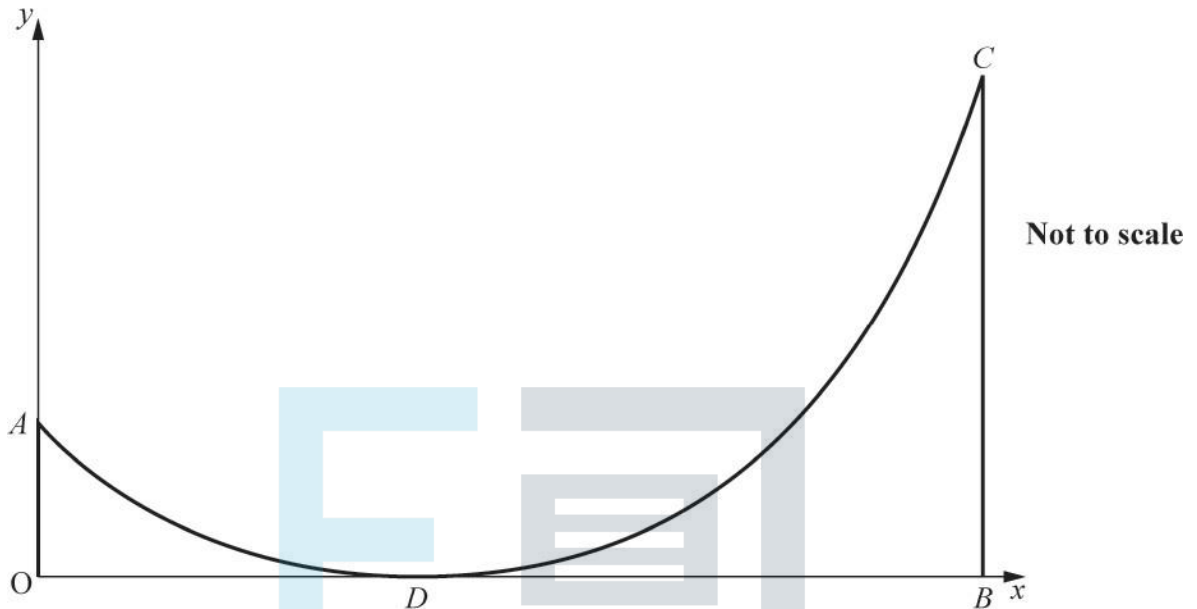
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6(a) Two thin poles, OA and BC , are fixed vertically on horizontal ground. A chain is fixed at A and C such that it touches the ground at point D as shown in the diagram.

On a coordinate system the coordinates of A , B and D are $(0, 3)$, $(5, 0)$ and $(2, 0)$.



It is required to find the height of pole BC by modelling the shape of the curve that the chain forms.

Jofra models the curve using the equation $y = k \cosh(ax - b) - 1$ where k , a and b are positive constants.

Determine the value of k .

[2]

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(b) Find the exact value of a and the exact value of b , giving your answers in logarithmic form.

[5]



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(c)

Holly models the curve using the equation $y = \frac{5}{4}x^2 - 3x + 3$.

Write down the coordinates of the point, (u, v) where u and v are both non-zero, at which the two models will agree.

[1]

(d) Show that Jofra's model and Holly's model disagree in their predictions of the height of pole BC by 3.32 m to 3 significant figures.

[3]

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7(a) Using exponentials, show that $\cosh 2u \equiv 2 \sinh^2 u + 1$.

[2]

(b) By differentiating both sides of the identity in part (a) with respect to u , show that $\sinh 2u \equiv 2 \sinh u \cosh u$.

[1]

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(c)

Use the substitution $x = \sinh^2 u$ to find $\int \sqrt{\frac{x}{x+1}} dx$. Give your answer in the form $a \sinh^{-1} b\sqrt{x} + f(x)$ where a and b are integers and $f(x)$ is a function to be determined.

[5]



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(d)

Hence determine the exact area of the region between the curve $y = \sqrt{\frac{x}{x+1}}$, the x -axis, the line $x = 1$ and the line $x = 2$. Give your answer in the form $p + q \ln r$ where p , q and r are numbers to be determined. [2]

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8(a)

The function $\operatorname{sech} x$ is defined by $\operatorname{sech} x = \frac{1}{\cosh x}$.

Show that $\operatorname{sech} x = \frac{2e^x}{e^{2x} + 1}$ [2]



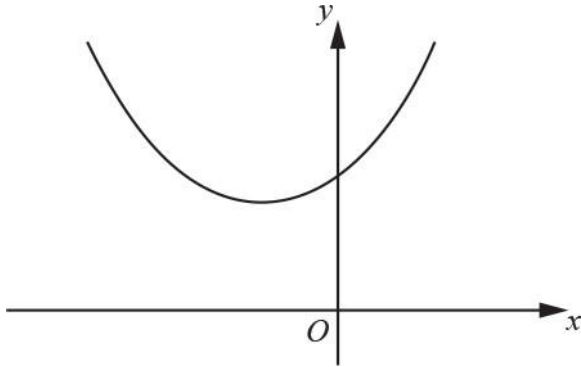
(b) Using a suitable substitution, find $\int \operatorname{sech} x \, dx$. [4]

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9(a) The diagram shows part of the curve $y = 5 \cosh x + 3 \sinh x$.



Solve the equation $5 \cosh x + 3 \sinh x = 4$ giving your solution in exact form.

[4]



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Find $\int_{-1}^1 (5 \cosh x + 3 \sinh x) dx$ giving your answer in the form $ae + \frac{b}{e}$ where a and b are integers to be determined.

[3]

- 10 In a predator-prey environment the population, at time t years, of predators is x and prey is y .
The populations of predators and prey are measured in hundreds.

The populations are modelled by the following simultaneous differential equations.

$$\frac{dx}{dt} = y \quad \frac{dy}{dt} = 2y - 5x$$

- (a) Show that $\frac{d^2x}{dt^2} = 2\frac{dx}{dt} - 5x$. [2]

- (b) (i) Find the general solution for x . [3]

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- (ii) Find the equivalent general solution for y . [2]

Initially there are 100 predators and 300 prey.

(c) Find the particular solutions for x and y .

[5]



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(d) Determine whether the model predicts that the predators will die out before the prey.

[4]



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- (a) Given that $u = \tanh x$, use the definition of $\tanh x$ in terms of exponentials to show that $x = \frac{1}{2} \ln \left(\frac{1+u}{1-u} \right)$. [4]



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- (b) Solve the equation $4 \tanh^2 x + \tanh x - 3 = 0$, giving the solution in the form $a \ln b$ where a and b are rational numbers to be determined.

[4]



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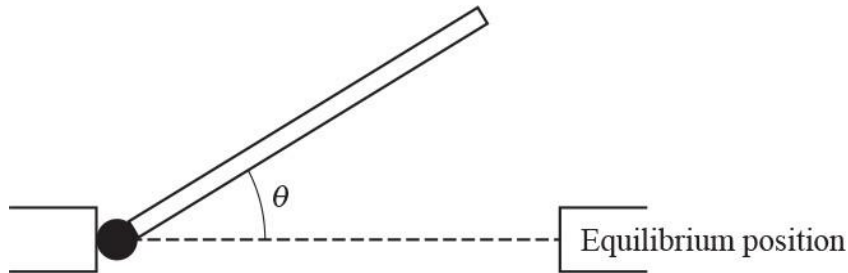
- (c) Explain why the equation in part (b) has only one root.

[1]

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- 12 A swing door is a door to a room which is closed when in equilibrium but which can be pushed open from either side and which can swing both ways, into or out of the room, and through the equilibrium position. The door is sprung so that when displaced from the equilibrium position it will swing back towards it.

The extent to which the door is open at any time, t seconds, is measured by the angle at the hinge, θ , which the plane of the door makes with the plane of the equilibrium position. See the diagram below.



In an initial model of the motion of a certain swing door it is suggested that it satisfies the following differential equation.

$$4 \frac{d^2\theta}{dt^2} + 25\theta = 0 \quad (*)$$

- (a) (i) Write down the general solution to (*).

[2]

- (ii) With reference to the behaviour of your solution in part (a)(i) explain briefly why the model using (*) is unlikely to be realistic. [1]

In an improved model of the motion of the door an extra term is introduced to the differential equation so that it becomes

$$4\frac{d^2\theta}{dt^2} + \lambda\frac{d\theta}{dt} + 25\theta = 0 \quad (\dagger)$$

where λ is a positive constant.

- (b) In the case where $\lambda = 16$ the door is held open at an angle of 0.9 radians and then released from rest at time $t = 0$.

Find, in a real form, the general solution of ($\dagger$). [3]

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(ii) Find the particular solution of (†).

[4]



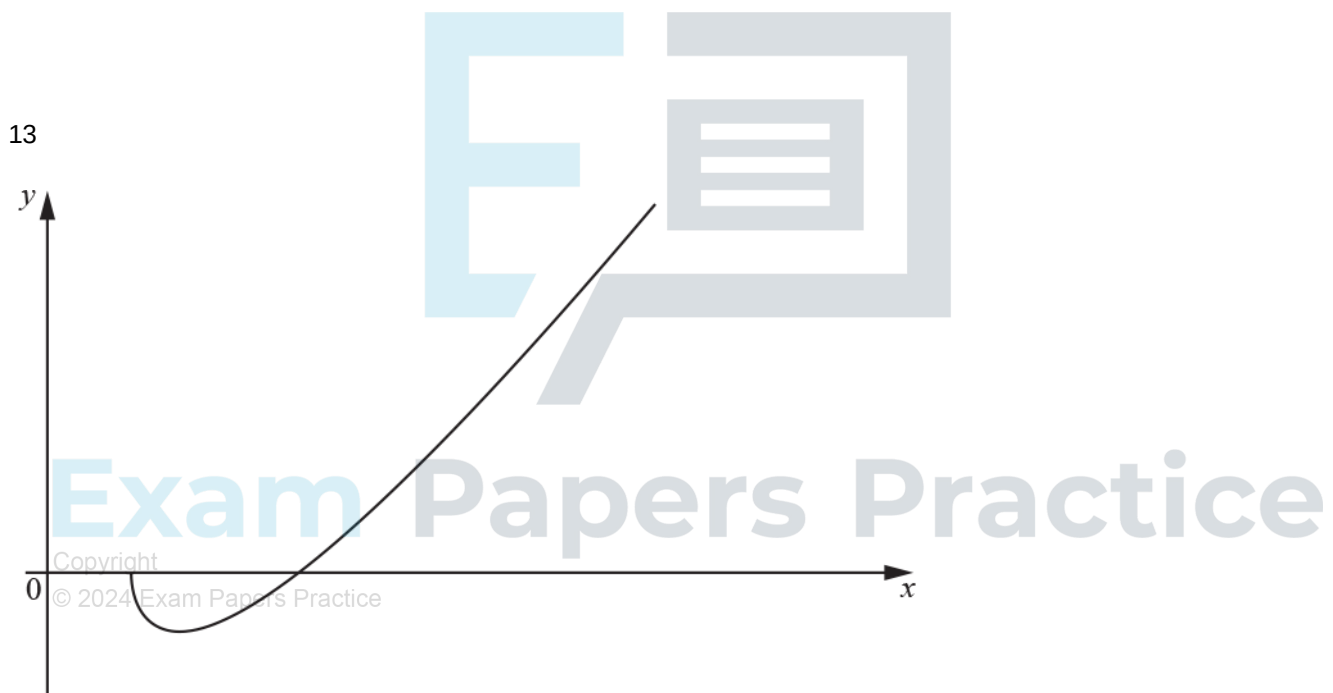
(iii) With reference to the behaviour of your solution found in part (b)(ii) explain briefly how the extra [2]
term in (†) improves the model.

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(c) Find the value of λ for which the door is critically damped.

[2]



The figure shows part of the graph of $y = (x-3)\sqrt{\ln x}$. The portion of the graph below the x -axis is rotated by 2π radians around the x -axis to form a solid of revolution, S .

Determine the exact volume of S .

[7]

- 14 (a) Using the definitions of $\cosh x$ and $\sinh x$ in terms of e^x and e^{-x} , show that $\sinh 2x = 2 \sinh x \cosh x$. [2]

You are given the function $f(x) = a \cosh x - \cosh 2x$, where a is a positive constant.

- (b) Verify that, for any value of a , the curve $y = f(x)$ has a stationary point on the y -axis. [2]

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- (c) Find the coordinates of the stationary point found in part (b). [1]

- (d) Determine the maximum value of a for which the stationary point found in part (b) is the only stationary point on the curve $y = f(x)$. [4]

You are given that for any value of a greater than the value found in part (d) there are three stationary points, the one found in part (b) and two others, one of which satisfies $x > 0$.

- (e) Find the coordinates of this point when $a = 6$.

Give your answer in the form $(\cosh^{-1} p, q)$. [4]

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- (a) Using the definition of $\sinh x$ in terms of e^x and e^{-x} , show that

$$4 \sinh^3 x = \sinh 3x - 3 \sinh x.$$

[3]

- (b) In this question you must show detailed reasoning.

By making a suitable substitution, find the real root of the equation

$$16u^3 + 12u = 3.$$

Give your answer in the form $\frac{a^{\frac{1}{b}} - a^{-\frac{1}{b}}}{c}$ where a , b and c are integers.

[5]

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16 In this question you must show detailed reasoning.

- (a) Find the coordinates of all stationary points on the graph of $y = 6\sinh^2 x - 13\cosh x$, giving your answers in an exact, simplified form. [9]



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- (b) By finding the second derivative, classify the stationary points found in part (a). [3]

- 17 (i) Using the definition of $\sinh x$ in terms of e^x and e^{-x} , show that $\sinh 3x = 4 \sinh^3 x + 3 \sinh x$. [3]

- (ii) Use the substitution $w = \sinh x$ to find the real root of the equation

$$4w^3 + 3w - 3 = 0.$$

Give your answer in exact form.

[4]



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- 18 Find the x-coordinate of the stationary point on the curve $y = 3 \cosh x - 2 \sinh x$, giving your answer exactly in logarithmic form. [3]

- 19 (a) Show that $\frac{d}{dx}(\sinh^{-1}(2x)) = \frac{2}{\sqrt{4x^2 + 1}}$. [2]

- (b) Find $\int \frac{1}{\sqrt{2 - 2x + x^2}} dx$. [3]

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The equation of a curve is $y = \cosh^2 x - 3\sinh x$. Show that $\left(\ln\left(\frac{3+\sqrt{13}}{2}\right), -\frac{5}{4}\right)$ is the only stationary point on the curve. [8]



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21 In this question you must show detailed reasoning.

The finite region R is enclosed by the curve with equation $y = \frac{8}{\sqrt{16+x^2}}$, the x -axis and the lines $x = 0$ and $x = 4$. Region R is rotated through 360° about the x -axis. Find the exact value of the volume generated. [4]

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(i) By first expanding $(e^x + e^{-x})^3$, or otherwise, show that $\cosh 3x \equiv 4 \cosh^3 x - 3 \cosh x$.

[4]

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(ii) Solve the equation $\cosh 3x = 6 \cosh x$, giving your answers in exact logarithmic form.

[5]



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23

By first expressing $\tanh y$ in terms of exponentials, prove that $\tanh^{-1}x = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$.

[3]



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24 It is given that $f(x) = 2 \sinh x + 3 \cosh x$.

(i) Show that the curve $y = f(x)$ has a stationary point at $x = -\frac{1}{2} \ln 5$ and find the value of y at this point.

[4]

(ii) Solve the equation $f(x) = 5$, giving your answers exactly.

[5]

25

By first completing the square, find $\int_0^1 \frac{1}{\sqrt{x^2 + 4x + 8}} dx$, giving your answer in an exact logarithmic form.

[6]

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(i) Given that $y = \cosh^{-1} x$, show that $y = \ln(x + \sqrt{x^2 - 1})$.

[4]

(ii) Show that $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2 - 1}}$.

[2]

(iii) Solve the equation $\cosh x = 3$, giving your answers in logarithmic form.

[3]

27

(i) Using the definitions for $\cosh x$ and $\sinh x$ in terms of e^x and e^{-x} , show that $\cosh^2 x - \sinh^2 x \equiv 1$.

[3]

(ii) Hence solve the equation $\sinh^2 x \equiv 5 \cosh x - 7$, giving your answers in logarithmic form.

[5]

28

It is given that $f(x) = \tanh^{-1}\left(\frac{1-x}{3+x}\right)$ for $x > -1$.

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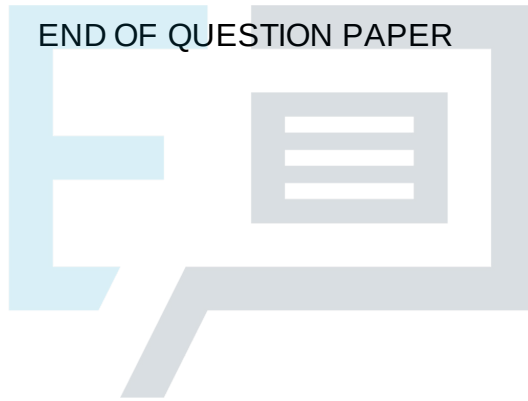
(i) Show that $f''(x) = \frac{1}{2(x+1)^2}$.

[6]

(ii) Hence find the Maclaurin series for $f(x)$ up to and including the term in x^2 .

[4]

END OF QUESTION PAPER



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